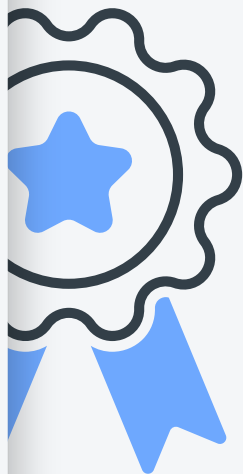
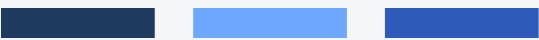


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Data Analysis

Transforming Data into Meaningful Insights



Overview

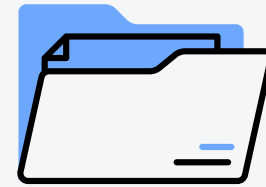
Over 31,000 UK train ticket transactions, covering payment method, ticket type/class, departure and arrival stations, journey dates and times, price, delay, and journey status (On Time/Delayed/Cancelled).



Excel

01

Exploration



03

Data Preparation



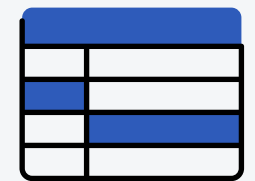
02

Calculations

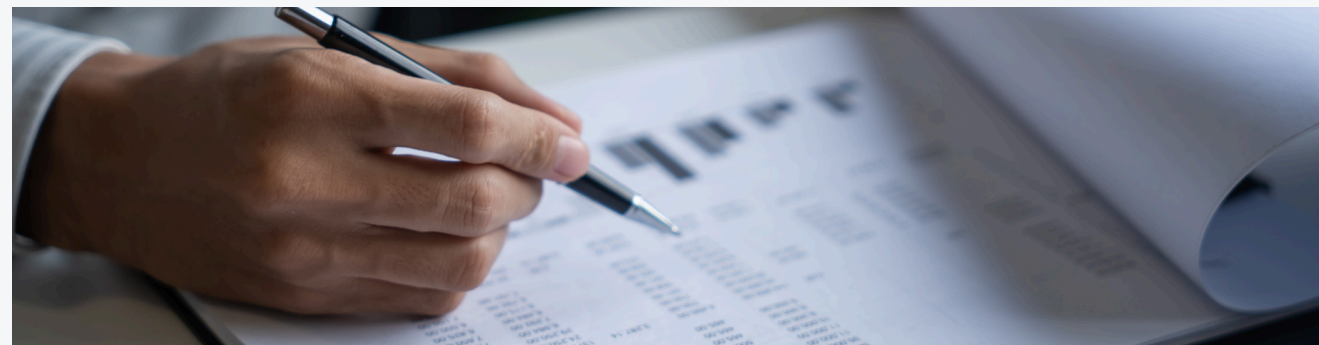


04

Visualization & Dashboard



Key Findings



Manchester had the highest trips (5,650 journeys)



Most trips were on time with very few delays



Refund requests were very low at just 4%

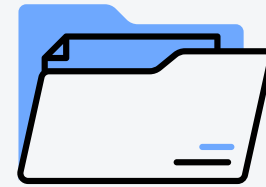


Standard class generated much higher revenue

Power BI

01

Exploration



03

Calculation of
Measures & Columns



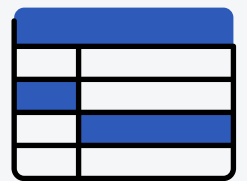
02

Solving Format Issues



04

Visualization



Key Findings



The most Used day is Wednesday.



Birmingham is a crucial station.



No. of tickets can be corelated with
educations times.

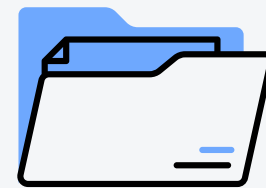


The online tickets is taking a good
share.

Tableau

01

Exploration



03

Finding the Insights



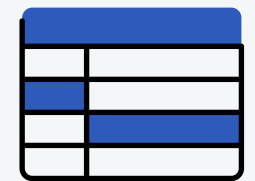
02

Data Cleaning

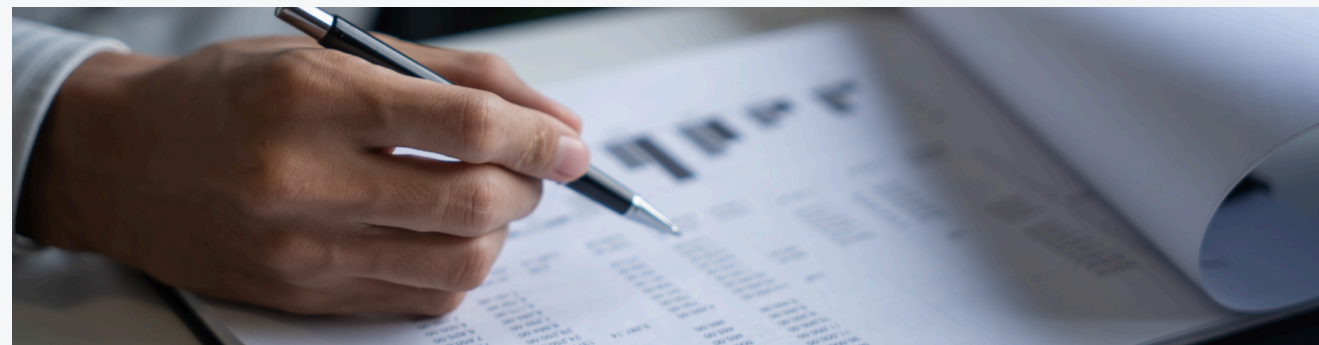


04

Building the Dashboard



Key Findings



Manchester Piccadilly is the busiest station.



Credit Card is the most used payment method.



Most trips run on time.

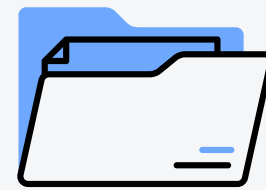


Manchester to Liverpool is the busiest route.

Python

01

Exeploration



03

المؤشرات الرئيسية (KPIs)

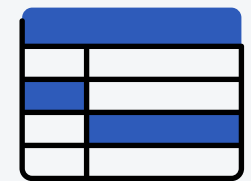


⁰²Data cleaning and prepartion



04

Visualizations



Key Findings



Revenue & Demand Stability



Booking & Discount Patterns



High On-Time Rate vs. Severe Delays

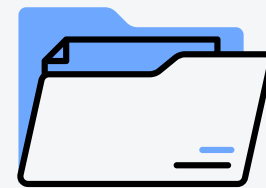


Route Concentration

Machine Learning

01

Data Quality



03

Best Model



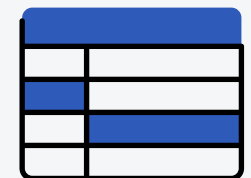
02

Data Leakage

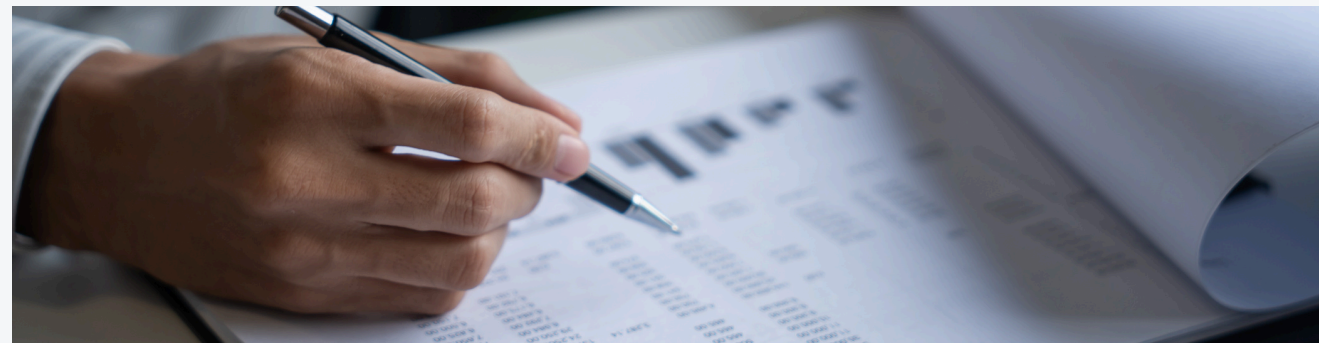


04

Important Features



Key Findings



Cleaned missing values, converted the target delay time into minutes, and encoded categorical features



Removed is_Delay and Journey Status because they could reveal the journey's actual delay outcome



Compared several regression models, and Random Forest achieved the best performance with $R^2 \approx 0.508$ and MAE ≈ 2.34 minutes.



Journey day, departure/arrival stations, purchase type, month, and price were among the most influential features

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Thank You

Questions & Discussion

